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CISC3003

Web Programming

Finalized Group Project Proposal

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20/04/2026

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Topic Name

The topic of our project is: “Mystery of Train Tracks: A Full-Stack Interactive Web Game”.

How can a responsive, secure, and feature-rich web application deliver both engaging gameplay and robust user services in a puzzle-based environment?

Purpose

Outcomes

This web minigame connects to the course learning outcomes by integrating front-end and back-end web development, database management, and user authentication. The project will enable us to enhance team collaboration and agile project management, which is essential for professional web application engineering. What is more, this project supports career planning by showcasing the ability to build scalable, secure and user-friendly applications. These skills obtained are relevant to roles in full-stack development, game design and system engineering.

Skills

This project will help develop skills that will prove to be useful for full-stack web development. They include the following:

- Front-end: Through responsive design with the usage of HTML, CSS and JavaScript technologies, to make the application work on desktop, tablet and smartphone devices.
- Back-end: Through usage of PHP and possibly other useful frameworks such as

Node.js and RESTful APIs, as well as usage of database systems, to facilitate players' data management.

- Authentication & Security: Through the implementation of login/signup, account verification, password reset and secure sessions.
- Search & Data Services: Designing the search capabilities for player's information and solving puzzles.
- Tracking & Analytics: The ability to monitor users' activity through integration of database logging capability.

Knowledges

As the project is dedicated to full-stack web development, it will allow us not only to gain necessary content knowledges, but also develop knowledges that we have learned in previous years of studies, which include the following:

- Database systems: Knowledge of schema design and querying, as well as relational and non-relational data storage techniques.
- Web security: Through the usage of login/signup capabilities, we will learn more about hashing, encryption and authentication.
- Game logic integration: Applying various algorithms for the game, and embedding puzzle mechanics into a web framework.
- Deployment: Once the game is complete without any errors, we host the website on cloud platforms.

Tasks

The following elaborates the tasks and deliverables for this project:

First of all, we start by planning and designing. For instance, we define the system architectures for front-end, back-end and database, then we create wireframes for the game's user interface (UI) and user account pages. For the database, we simply draft some E-R diagrams for user accounts, game records and visiting history.

Once the planning is done and agreed by our members, we start implementing in three different aspects, and the task distribution is as follows:

- Bruce and Collins will build a responsive UI for the puzzle game and user dashboard.
- John and Uriel will be responsible for implementing APIs for authentication, search and history tracking. They will also implement the signup/login system with email verifications, as well as adding password reset functionality.
- As for Seki and Joaquin, their work will revolve around the database systems; this involves creating schemas for users, game logs, puzzle solutions, etc. They will also be responsible for coming up with the search system of player logs and solutions to puzzles, as well as recording the history of visits with dates and URLs.

Lastly, we will do unit testing and usability testing among our peers. Once the testing is complete with no errors, we will deploy the application to cloud hosting service.

The following requirements will guide us throughout the development of our project.

The website we are expecting:

- The website needs to be built using responsive design so that the game can function properly on all devices.
- The user interface allows the player to play the game smoothly.
- The application has a login and registration feature.
- The complete website must be deployed and accessible online.

Our project consists of the following deliverables:

- Fully functional Train Puzzle Game

- Secure login/signup system with email verification
- Search service for records and solutions
- Visiting history and tracking
- Documentation (e.g. README, system design, user guide)
- Demo gameplay

Tools

The tools we will use during the development of the train puzzle game are as follows:

Firstly, for front-end interactive design, we use the standard technologies such as HTML, CSS and JavaScript. Especially for game logic and user interactions, JavaScript would be dominant throughout the development of the game.

Secondly, we will use both Node.js and PHP for back-end development. PHP integrates seamlessly with MySQL and phpMyAdmin, making it suitable for implementing core functionalities such as user registration, login, and account activation. Node.js, on the other hand, features asynchronous non-blocking I/O, making it suitable for handling high-concurrency requests and facilitating future expansion into more complex API services or real-time interactive functions.

Thirdly, for database management, MySQL would be our primary choice. In terms of strict data structure, MySQL is a relational database that enforces a predefined schema. For instance, every row must adhere to specific data types and constraints. This prevents "dirty data" from ever being saved. Regarding complex analytical queries, MySQL uses standard SQL, which is purpose-built for handling intricate data relationships. Looking at MongoDB which offers \$lookup, it is often less efficient and more cumbersome for deeply nested or multi-step relational analysis.

Lastly, there are supplement tools for us to deploy our web application properly without

any exceptional errors. Deployment platforms such as Heroku, AWS and Vercel are some of the most common choices for developers to introduce their products to the Internet. GitHub ensures smooth, seamless deployment primarily through its automated workflow engine and integrated hosting services.

Our Workflow Plans

Front-end Interactive Design

The main components in the game include the puzzle itself, the game menu and the players' result. The first objective is to design the game components such as tracks, blank space, trains. These parts can be designed using 3D blending applications or simply finding them via online assets. After making these components, we then create a simple level and work on game logic and algorithms. For instance, the objective of the game is to move tracks so that the train can move from one tunnel to another. Once the basics of the level is implemented smoothly, we then add more functions such as adding more levels, as well as features of menu and results.

Back-end Development

The back-end for this project primarily focuses on developing a login/signup system for players who are interested in saving their game records. Designing an engaging, game-themed authentication interface utilizing HTML, CSS and JavaScript ensures a seamless and responsive user experience. For the server side, PHP is applied to handle critical operations such as session management, secure password modification and automated email dispatching. Once the system is done, we will establish a secure connection with the database system, where all user passwords will be safeguarded using hashing algorithms to ensure data privacy and security.

Database Management

Creating a database system is crucial for the game and players who are willing to save their game records. We start by defining database schemas for core entities such as user,

player profile, records and visiting log. Then we start integrating with the game interface. For instance, the database system will interact with the game's front-end to save records, update scores, retrieve profiles and query game history. A logging system is built to track user visits such as login IP, device information and timestamps. This way, the database system will support user authentication and data queries, ensuring secure and efficient access to players and their game data.

Possible Challenges

We expect challenges such as integrating the game logic into the full-stack web framework smoothly, ensuring secure user authentication, and managing real-time history tracking with no performance issues. What is more, we encounter questions from different teams, making us think further when we are developing the game.

Since we start with working each aspect by separate pairs, an agile approach will be adopted where there will be a well-defined distribution of duties among all members. The front-end developers will concentrate on creating user-friendly UI components, whereas the back-end programmers will work towards designing APIs that provide well-organized endpoints. The database programmers will work towards ensuring consistency in schemas and optimizing the queries involved. Continuous sprint meetings and integration testing will help in spotting mismatches between different system tiers.

In a certain case if our system scales up, we will employ a layered architecture and load balancing to distribute request pressure, while using indexes, caching, and partitioning at the database level to maintain query performance. The frontend enhances user experience through reactive design and asynchronous requests, while the backend can be scaled to a microservice architecture as needed to ensure performance and responsiveness under high concurrency.

As the number of users grows, we will employ efficient indexing mechanisms, caching

strategies, and, when necessary, separate player data from personal profiles. At the same time, users will have data autonomy, able to delete their browsing history in their personal settings to ensure privacy.

Criteria for Success

A successful final product should possess the following characteristics:

- The website loads completely without JavaScript or CSS errors.
- The web must consist of responsive design, ensuring the game displays well on any devices.
- The UI provides smooth gameplay with intuitive controls.
- The application includes a secure login/signup system with email verification and password reset.
- The webpage offers search functionality for player records and puzzle solutions.
- The web game tracks visiting history with clear logs accessible to users.
- Outside of the website, there should be documentation such as README, annotated examples and, if possible, a demo gameplay.
- The project is successfully deployed on a PHP-enabled platform.

The scoring criteria are as follows:

- Content accuracy and relevance (35%)
- Technical implementation and functionality (35%)
- Design and user experience (20%)
- Documentation completeness (10%)

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